Noetic Diffusion Theory: A Rhythmic, Geometric Framework for Neural State Reconstruction and Conscious Correlates

Version 0.8 · 10.5281/zenodo.20112950 · August 10, 2026

Robin Langell
Independent Researcher
For correspondence: Robin Langell — hello(at)noeticdiffusion.com

Abstract

We present Noetic Diffusion Theory (NDT) as a testable framework for neural state dynamics: conscious-state correlates are modeled as time-resolved denoising on a learned geometric manifold. The Meta–Noetic Diffusion Model (MNDM) formalizes this process as a guided stochastic differential equation with a rhythmic variance schedule, and Rhythmic Variance Control (RVC) links that schedule to measurable neural rhythms. The core empirical target is not a complete theory of subjective experience, but a constrained set of dynamical signatures that can be tested in neural recordings.

To make the framework measurable, we define the Meta–Noetic Phase Space (MNPS) as the canonical Stage-1 coordinate system; optional Noetic Atlas readouts are specified in Appendix Section 16.2. The first-stage analysis focuses on three primary claims: event-locked entropy changes around slow-oscillation–spindle complexes, sleep-stage transformations in MNPS topology (NREM centralization and REM expansion), and bounded disorder-related geometry contrasts that must survive preregistered multiverse controls. A composite Noetic Diffusion Health Index (NDHI) is retained only as an exploratory visualization and summary index, not as a clinical biomarker or consciousness score.

The article therefore offers a staged research program: a minimal dynamical model, a reproducible measurement layer, explicit null models, and falsifiable public-dataset tests. Speculative extensions—embodied anchoring, detailed TRN biophysics, landscape learning, second-order Jacobian analyses, reachability, and clinical translation—are treated as appendices or follow-on work rather than prerequisites for the core NDT claim.

1. Scope and Epistemic Status

NDT provides a mechanistic framework for neural dynamics associated with conscious-state reports, but we carefully distinguish three levels of claim:

Strong claims (empirically testable):

- Neural rhythms (theta–gamma PAC, SO–spindle coupling) implement variance scheduling that produces measurable entropy dynamics.
- Manifold topology in Meta–Noetic Phase Space (MNPS) systematically varies with sleep stage, task context, and some clinical contrasts.
- Null models without event-locked variance scheduling should fail to reproduce the strongest rhythmic and topological signatures.

Moderate claims (hypothesized correlations):

- These neural geometric signatures correlate with reported conscious states (presence, clarity, embodiment).
- First-person phenomenology is hypothesized to exhibit structure (pulsed quality, discrete epochs) aligned with neural rhythmic events.
- Clinical groups may show characteristic manifold deformations, but these are treated as bounded, dataset-specific hypotheses rather than diagnostic categories.

Speculative claims (philosophical):

- The geometric flow described by MNDM may be constitutive of consciousness (experience is the intrinsic character of rhythmic denoising).
- This constitutive hypothesis does not generate empirical predictions beyond those of the correlational dynamical model; it is a metaphysical stance on what the already-specified denoising dynamics amount to. All concrete predictions in this manuscript bear on the presence or absence of the denoising/geometric signatures themselves; if those are supported, one may choose to read them either in a purely correlational or in a more constitutive spirit, but no behavioural or neural test uniquely adjudicates between these readings.

Extension-status rule. Later NDT work includes synthetic recoverability audits, subsystem hypotheses, and measurement-layer companion analyses. These have distinct evidentiary roles: a truth-known simulator can validate an estimator under its frozen synthetic contract without validating an empirical MNJ; an observational body–brain association can constrain EAP without identifying causal entrainment; and a circuit proposal remains a hypothesis until its own causal test succeeds. We do not promote any of these extensions to the Stage-1 claim set by inclusion in this article.

Throughout, we use "conscious state reconstruction" as shorthand for "neural dynamics exhibiting systematic correlates with conscious states." The hard problem of why any neural process is accompanied by subjective experience remains unaddressed.

2. Introduction

This paper introduces a formal, testable account of how neural systems may rhythmically alternate between exploration and denoising, and how those dynamics become visible as geometric structure in neural state space. We use Noetic Diffusion Theory (NDT) as the scientific umbrella, the Meta–Noetic Diffusion Model (MNDM) as the mathematical core, and the Meta–Noetic Phase Space (MNPS) as the measurement layer. Here, "noetic" denotes knowledge-bearing or intelligible structure: latent neural states whose organization can be measured in a principled way. Our framing complements predictive coding/active inference perspectives [1], [2] and draws on neurophilosophical accounts of state-space representation [3], [4], [5] and real patterns [6].

> **When you wake from a dream, the world reconstructs itself in seconds. Chaotic imagery resolves into your bedroom; formless unease crystallizes into the memory of today's deadline. This transition—from noise to meaning—is neither instantaneous nor continuous, but iterative, rhythmic, and geometrically structured.**

The motivating phenomenological intuition is that some conscious episodes may feel like progressive reconstruction: vague, noisy structure becomes a usable world model over short time windows. This intuition motivates the language of denoising, but the core tests in this manuscript are third-person dynamical tests: event-locked entropy, MNPS topology, recurrence, and model comparison against non-rhythmic nulls. First-person anchoring, embodiment, and high- resolution neurophenomenology are treated as Tier-2 extensions rather than requirements for the initial validation of NDT.

Reading guide. Readers mainly interested in the formal core can focus on Section "The Meta–Noetic Diffusion Model (MNDM)", the RVC section, and the Predictions/Identifiability sections. Readers planning empirical analyses should prioritize the MNPS methods and public-dataset plan. The appendix collects extensions that are useful for later work but not required for the core claim: embodied anchoring, TRN biophysical detail, landscape learning, and related theory comparisons.

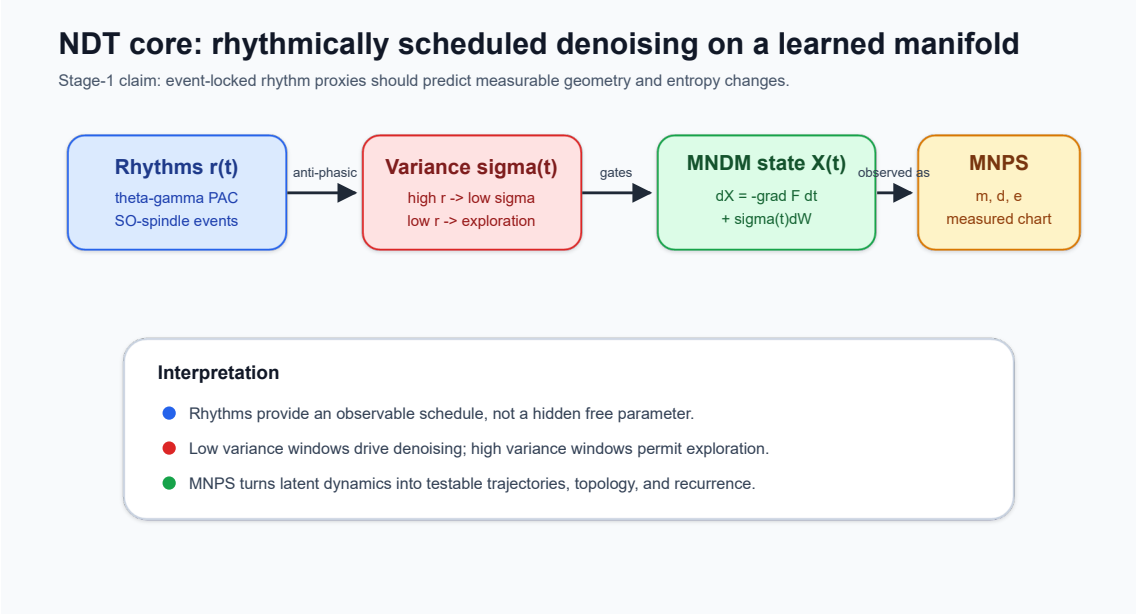


Figure 1: Core NDT flow. Rhythmic control $r(t)$ modulates the variance schedule $\sigma(t)$, which gates denoising and exploration in the MNDM state trajectory. MNPS then supplies the Stage-1 measurement chart used for empirical tests.

3. The Meta–Noetic Diffusion Model (MNDM)

Intuition. The brain maintains a noetic state X that balances exploration (maintaining uncertainty to sample alternatives, consider possibilities) against coherence (collapsing uncertainty into stable, actionable meaning). Too much exploration yields confusion; too much coherence yields rigidity. Rhythms set when to explore vs. cohere, implementing a dynamic schedule that navigates this trade-off. The MNDM formalizes this as a guided diffusion on a learned potential landscape F (acquired through development and experience): trajectories descend where meaning is strong, pause or re-explore when rhythms loosen variance, and re-stabilize when variance tightens. We now formalize this intuition mathematically, beginning with a minimal system capturing the essential dynamics.

3.1. Minimal formulation

We begin with a minimal, citable system capturing the essence of time-resolved denoising modulated by rhythm before introducing the full extended potential. Three ingredients define it: a diffusion equation for the noetic state, a potential trading off disorder against coherence, and a rhythm-gated variance schedule.

State update. The noetic state X_t descends a potential F under rhythm-gated noise:

$$dX_t = -\nabla F(X_t, t)dt + \sigma(t)dW_t \tag{1}$$

Minimal potential. F trades off disorder against coherence, using the same (m, d, e) coordinates as the MNPS measurement chart (see the notation snapshot below):

$$F_{\min(X)} = e(X) - m(X) \tag{2}$$

Variance schedule. Rhythmic control $r(t) \in [0, 1]$ gates the diffusion variance between a floor $\sigma_{\min}$ and a ceiling $\sigma_{\min} + \sigma_0$:

$$\sigma(t) = \sigma_{\min} + \sigma_0(1 - r(t)) \tag{3}$$

Here e measures disorder (for example, entropic complexity), m measures rhythmic coherence/structure, and W_t is Brownian motion (modeling fast fluctuations as isotropic noise for tractability, though biological noise is likely colored and signal-dependent). In this minimal case, F has no explicit time dependence (though X_t evolves); the extended formulation (Eq. 4) will add time-varying schedules.

The minimal SDE mirrors score-based diffusion formalisms in machine learning [7], [8], adapted here with a biologically grounded variance schedule.

Notation snapshot (minimal vs. extended potential and MNPS axes). Throughout the manuscript we keep the MNPS axes in the unified notation (m, d, e), where e is an entropy/disorder coordinate, m is a mobility / rhythmic-coherence coordinate, and d is a diffusivity / deviation-from-metastability coordinate. In older drafts these same feature-families were denoted (E, R, D); here we use (m, d, e) throughout for consistency across manuscripts.

We use two closely related forms of the noetic potential: $F_{\min(X)} = e(X) - m(X)$ in the minimal model (Eq. 2) and $F(X,t) = \alpha(t)e(X) - \beta(t)m(X) + \gamma(t)d(X) - \delta(t)\Phi(t)$ in the extended model (Eq. 7). The 3D MNPS chart (m, d, e) is defined at the measurement layer irrespective of which terms currently enter the potential; in the strict minimal MNDM we can view the dynamics as evolving in the (m, e) subspace (with $\gamma = 0$) while still tracking d as an additional diagnostic coordinate. When we compare minimal and extended models empirically, we therefore treat the minimal model as a 2D projection of the full (m, d, e) manifold that ignores feedback from d into F but allows its geometry to be measured.

Intuitively, when rhythmic control $r(t)$ is **high** (for example, during a theta–gamma burst or SO–spindle complex), variance $\sigma(t)$ is **low**, making denoising aggressive—the system descends F rapidly. When $r(t)$ is **low** (between rhythmic events or during REM), variance $\sigma(t)$ is **high**, enabling exploration—the system can climb F to escape poor local minima or sample alternatives. This anti-phasic relationship (high r versus low σ) is the core of rhythmic variance control, conceptually analogous to precision scheduling in inference and noise schedules in diffusion models [9].

Limiting cases. If $r(t) \equiv 1$ (constant maximal rhythm), $\sigma(t) \to \sigma_{\min}$ and the system approaches (near-)deterministic gradient descent on F. If $r(t) \equiv 0$ (no rhythmic control), $\sigma(t) = \sigma_{\min} + \sigma_0$ is maximal and the system performs largely unguided exploration. Healthy cognition requires $r(t)$ to **vary**, alternating denoising and exploration.

3.1.1. Note on pathological extremes (the Grand Mal boundary)

A critical limitation of the minimal potential $F_{\min} = e - m$ is that it can mistake low entropy plus high rhythmic structure for successful denoising. Taken to its logical extreme, a generalized tonic-clonic seizure—with hypersynchronous high-amplitude oscillations and reduced signal complexity—could look like a maximally coherent state. This is phenomenologically false and clinically absurd.

The resolution is to treat the minimal model as a **physiological-corridor** approximation, not as a complete regime classifier. As Churchland (1986, 1995) emphasizes, awareness requires functional differentiation as well as integration [3], [5]. Later NDT work makes the same point operationally: reduced capacity must be decomposed into capacity collapse, transport-innovation balance, and pathological lock-in rather than collapsed into a single scalar [10]. A seizure-like regime is not merely "low entropy"; it is transport-captured, canalized, and pathologically locked.

The extended potential therefore includes the deviation term $d(X)$ (Eq. 7), which penalizes departures from an integration-segregation corridor. In the full model, meaningful coherence is metastable and traversable; pathological hypersynchrony is a degenerate attractor outside the valid operating range of Minimal-MNDM.

Term-by-term interpretation.

Symbol	Neurobiological interpretation	Empirical signature
X_t	Noetic state (latent neural/phenomenal configuration)	High-dimensional neural activity (EEG/fMRI)

Symbol	Neurobiological interpretation	Empirical signature
$e(X)$	Disorder/uncertainty	Spectral entropy, sample entropy, Lempel–Ziv complexity
$m(X)$	Mobility / rhythmic coherence/structure	PAC, PLV/wPLI, spectral power in bands
$F(X)$	Noetic potential ($e - m$)	Composite: e minus normalized m
$r(t)$	Rhythmic control (theta, SO–spindle)	Power envelope of theta/SO, spindle density
$\sigma(t)$	Effective variance schedule	$\sigma_0(1 - r(t))$, estimated from $r(t)$
W_t	Fast stochastic fluctuations	High-frequency broadband noise

MNDM variables with neurobiological interpretation and measurable signatures.

Note on scales. e and m are normalized to comparable ranges (for example, $[0, 1]$) so their difference $F = e - m$ is well-defined. The specific normalization is part of MNPS coordinate construction discussed later in the measurement section.

Assumptions and invariances.

- **Coordinate choice:** The features $(e, m, ...)$ defining MNPS are **charts** on the underlying noetic manifold—local coordinate systems. Different choices (for example, different entropy measures, PAC vs. PLV for m) define different charts. Our predictions concern **intrinsic geometric properties** (trajectory curvature, geodesic distance, manifold topology) that are invariant under smooth coordinate transformations. This allows empirical flexibility while maintaining theoretical precision.
- **Monotone transforms:** Replacing e or m with monotone equivalents preserves qualitative flows (order relations matter, not absolute scales).
- **Discrete vs. continuous dynamics:** While the SDE formalism is formally continuous time, the **biological implementation** via rhythms creates **discrete denoising epochs** (theta cycles, spindle events). This discreteness is essential to NDT's step-wise predictions and distinguishes it from continuous-variance frameworks. While standard ML diffusion solvers (e.g., Euler-Maruyama) are numerically discrete, NDT posits that the discreteness is **biologically imposed** by the rhythm $r(t)$ itself, not merely an artifact of integration steps [7], [8].
- **State invariance:** The MNDM form (Eq. 1) applies across wake, NREM, and REM; only the **schedules** $r(t)$ and the landscape weights change. This invariance allows us to use the same measurement framework (MNPS) across states while predicting state-specific geometry.

3.2. Observation model and score estimation

We explicitly distinguish latent noetic state dynamics from measurements. Let

$Y_t = g(X_t) + \varepsilon_t,$

where $g : \mathbb{R}^d \to \mathbb{R}^p$ is a locally C^1 observation map (for example, from latent state to multichannel EEG/fMRI) and ε_t is zero-mean, finite-variance noise. The MNPS coordinates (m, d, e) are computed from short windows of Y_t; differentiability of g ensures that, under the chain rule and mild regularity of feature functionals, gradients with respect to X exist almost everywhere.

In empirical settings $\nabla F(X, t)$ is not available in closed form. We therefore estimate it via one of two routes:

1. **Direct potential parameterization:** Posit $F(X, t; \theta)$ and learn θ by fitting trajectory statistics or via contrastive/energy-based objectives on MNPS embeddings; use $\nabla_X F(X, t; \theta)$ during simulation.
2. **Score-based route:** Learn a score field $s_{\varphi(X,t)} \approx \nabla_X \log p_{t(X)}$ (for example, via score matching on MNPS embeddings of Y_t [7], [9]), and set $\nabla F(X, t) \propto -s_{\varphi(X,t)}$ up to state-dependent rescaling

that enforces the $e - m$ structure. This identifies the **direction** of descent without requiring closed-form F.

Assumptions used throughout: (i) g is locally Lipschitz; (ii) feature maps used to build MNPS are Fréchet-differentiable almost everywhere (or **weakly differentiable** in the distributional sense); (iii) empirical gradients are estimated in the MNPS chart, which we use as coordinates for X when learning from data. In practice, the score-based route relies on smoothness **in distribution** rather than pointwise differentiability, aided by temporal windowing and regularization.

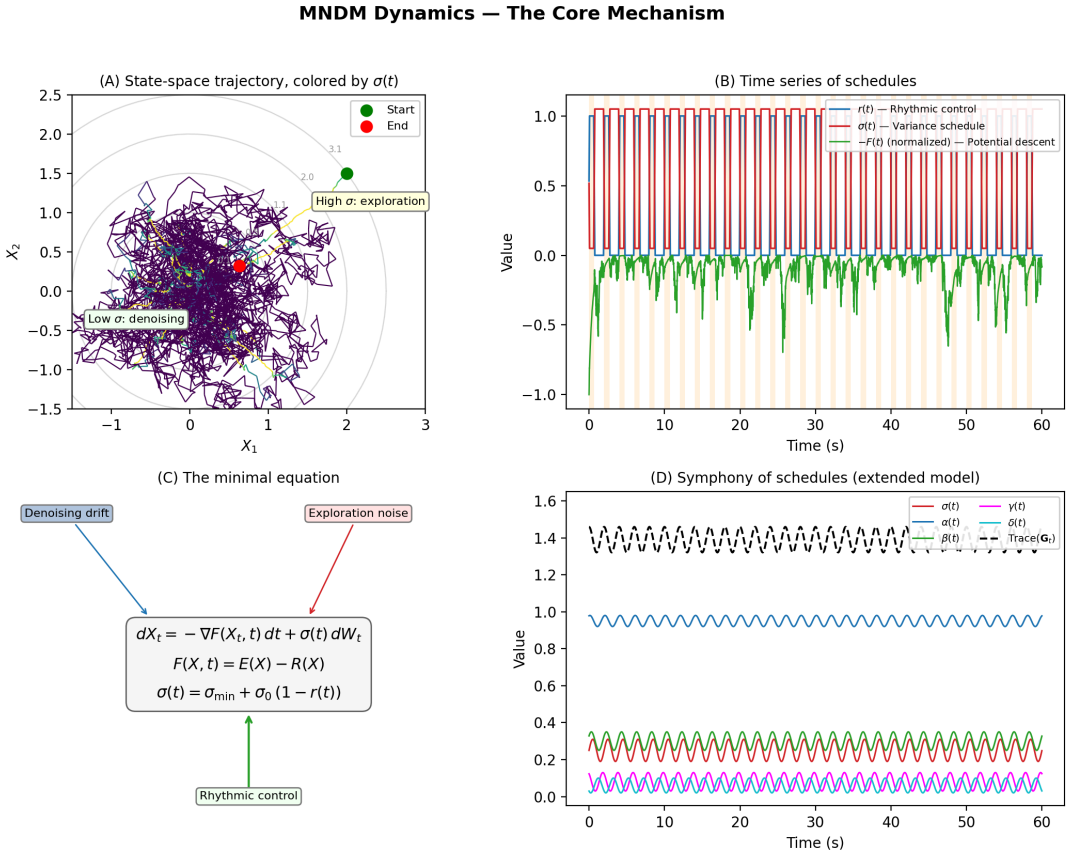


Figure 2: (Simulated) MNDM dynamics with rhythmic variance control, generated by `src/make_figures.py`. **A** State-space trajectory of the minimal model, colored by the local variance schedule $\sigma(t)$; contour lines represent the potential landscape $F(X)$. **B** Anti-phasic relationship between rhythmic control $r(t)$ (blue pulses) and variance schedule $\sigma(t)$ (red, inverse relationship), with the potential's descent (green) tracking low-variance denoising epochs (shaded). This panel directly visualizes Eq. 1. **C** The minimal MNDM equation with visual annotations showing the interplay between denoising drift, exploration noise, and rhythmic control. **D** Coordination of schedules in the extended formulation (Eq. 4): illustrative evolution of landscape weights $(\alpha, \beta, \gamma, \delta)$, with Trace($\boldsymbol{G}_t$) summarizing the attention geometry tensor.

3.3. Extended formulation (geometry and meta-field)

Why extend the minimal model? The minimal formulation (Eq. 1) assumes uniform attention across the noetic manifold and fixed balance between entropy and coherence. But the brain clearly modulates **what** to denoise (attention) and **when** different pressures dominate (varying schedules).

We hypothesize that the **neural correlates of consciousness** are orchestrated by multiple coordinated schedules operating at different timescales—variance $\sigma(t)$ setting the tempo, landscape weights $\alpha, \beta, \gamma, \delta$ composing the score, geometry $\boldsymbol{G}_t$ directing attention, and embodied anchoring providing the somatic foundation. The extended formulation formalizes this multi-schedule architecture, adding three capabilities and linking to metastability/criticality regimes [11], [12], [13]:

1. **Adaptive geometry** ($\boldsymbol{G}_t$): Attention reshapes the manifold, stretching some dimensions (amplifying relevant features) and compressing others.
2. **Time-varying schedules** ($\alpha, \beta, \gamma, \delta$): Different moments prioritize entropy reduction, coherence, homeostasis, or meta-regulation differently.
3. **Meta-field** (Φ): A slow regulatory variable that adapts based on how well denoising is working—a form of "confidence" or "meta-cognitive monitoring".

We now extend Eq. 1 to incorporate these mechanisms. The extended system adds four ingredients to the minimal state update: a geometry tensor that reshapes the descent direction, an adaptation rule for that geometry, an adaptation rule for the meta-field, and a richer, time-varying potential. The variance schedule keeps the same form as Eq. 3.

State update with adaptive geometry. The descent direction is now reshaped by a state-dependent metric $\boldsymbol{G}_t$:

$$dX_t = -\boldsymbol{G}_{t(X_t)} \nabla F(X_t, t) dt + \sigma(t) dW_t \tag{4}$$

Geometry adaptation. $\boldsymbol{G}_t$ evolves slowly in response to the local gradient, the meta-field, and rhythmic control:

$$\dot{\boldsymbol{G}}_t = \mathcal{A}(\boldsymbol{G}_t, \nabla F, \Phi, r(t)) \tag{5}$$

Meta-field adaptation. The meta-field Φ tracks how well denoising is currently working, via the metric-weighted gradient norm:

$$\dot{\Phi} = \mathcal{B}(\Phi, \langle \nabla F, \boldsymbol{G}_t \nabla F \rangle, r(t)) \tag{6}$$

Extended potential. Landscape weights $(\alpha, \beta, \gamma, \delta)$ are now allowed to vary in time and to depend on the meta-field:

$$F(X, t) = \alpha(t) e(X) - \beta(t) m(X) + \gamma(t) d(X) - \delta(t) \Phi(t) \tag{7}$$

Here $\boldsymbol{G}_t$ is a **state-dependent Riemannian metric tensor** on the noetic manifold—essentially a positive-definite matrix that defines local distances and determines how gradient descent operates. It represents **attention** in geometric terms: directions with large eigenvalues are "attended to" (steeper descent), directions with small eigenvalues are "ignored" (shallow descent). Connections to manifold learning and information geometry provide complementary estimation tools [14].

3.4. Well-posedness and stability

To guarantee existence/uniqueness and numerical stability, we assume:

- **(A1) Lipschitz/linear growth:** $\nabla_X F(\cdot, t)$ is globally Lipschitz with at most linear growth uniformly in t; schedules $\alpha, \beta, \gamma, \delta$ are piecewise-C^1 and bounded.
- **(A2) Non-degenerate diffusion:** $\sigma(t) \in [\sigma_{\min}, \sigma_{\max}]$ with $\sigma_{\min} > 0$, piecewise continuous.
- **(A3) SPD geometry:** $\boldsymbol{G}_t$ is symmetric positive-definite with eigenvalues in $[\lambda_{\min}, \lambda_{\max}]$ and dynamics $\mathcal{A}$ preserve SPD.
- **(A4) Timescale separation:** To prevent overfitting where geometry chases noise, we assume $\| \dot{\boldsymbol{G}}_t \| \ll \| \dot{X}_t \|$ on the timescale of single denoising steps. $\boldsymbol{G}_t$ evolves as a slow variable (plasticity) relative to the fast state dynamics X_t.
- **(A5) Regular meta-dynamics:** $\mathcal{A}$ and $\mathcal{B}$ are locally Lipschitz in their arguments.

Under (A1)–(A5) the SDE in Eq. 4 admits a unique strong solution and preserves $\boldsymbol{G}_t \succ 0$; $\sigma_{\min} > 0$ avoids degeneracy in long intervals. These conditions anchor empirical estimation and simulation choices.

New terms in the potential.

- $d(X)$: Deviation from optimal metastability (for example, $|Q - Q^*|$ where Q is integration/segregation balance). Penalizes extreme states (too integrated or too segregated). **Crucially, this term prevents the landscape from directing the system toward pathological hypersynchrony (seizures), distinguishing "meaningful" structure from "hyper-ordered" collapse.**

- $\Phi(t)$: Meta-regulatory field, increases when denoising is effective (low $\langle \nabla F, \boldsymbol{G} \nabla F \rangle$), decreases otherwise. Acts as adaptive damping.
- Schedules $\alpha, \beta, \gamma, \delta$: Time-varying weights. Example: During focused tasks, α and β increase (entropy reduction and coherence priority); during REM, α and β decrease (explore freely), while δ increases (meta-regulation dominant).

3.5. Coherence and quasi-stability

We define a **coherence functional**

$$\Omega(X, t) = e^{-F(X,t)} \tag{8}$$

which increases as the system achieves lower potential F.

Note on monotonicity: Since F typically includes the deviation term $d(X)$ (Eq. 7), minimizing F does not imply seeking a "dead," static equilibrium. Rather, it implies seeking the **optimal balance** defined by the landscape (e.g., critical integration-segregation). Thus, high coherence Ω reflects proximity to this optimal metastable regime, not thermodynamic freeze.

We emphasize that Ω is **not** a Lyapunov function in the strict sense—we do not claim monotonic increase—because:

1. Stochasticity $\sigma(t)$ allows uphill movements (exploration).
2. Time-varying schedules $\alpha(t), \beta(t), \gamma(t), \delta(t)$ can increase F temporarily.
3. Adaptive geometry $\boldsymbol{G}_t$ evolves, changing the landscape.

Instead, Ω serves as an **instantaneous coherence measure**: Higher Ω indicates a currently more meaningful state. Over a denoising epoch (a window where σ decreases), we expect **cycle-averaged coherence**

$$|\Omega|_T = \left(\frac{1}{T}\right) \int_0^T \Omega(X_t, t) dt \tag{9}$$

to increase on average. This quasi-stability—**stability on average with fluctuations**—is the hallmark of healthy cognitive dynamics.

Reporting and normalization. Because Ω can span orders of magnitude, for empirical plots we typically work with a stabilized and normalized version, such as the within-subject z-score of $\log \Omega(X, t)$, and equivalently a normalized potential $\tilde{F}$ when comparing groups. This enforces commensurate scales for e, m, and d and avoids unit-driven dominance.

Pathological extremes. Excessive stability (Ω always high, F always low) creates rigid cognition; excessive instability (Ω highly volatile, F flat) creates chaotic cognition. The framework predicts an **optimal regime** of moderate Ω variability around a healthy mean.

3.6. Model variants for Tier-1 testing

For the first empirical wave we use Minimal-MNDM as the default model: global (m, d, e) MNPS coordinates, fixed geometry, no meta-field, and a rhythm-derived variance schedule. The main comparator is a continuous-precision/rhythm-smoothed null, which tests whether event-locked variance is actually needed to explain step-wise entropy signatures.

More elaborate variants—fixed-geometry, no-meta, continuous-rhythm, and extended-geometry forms—are part of the model-comparison toolkit but are moved to Appendix Section 16.1. The practical rule is hierarchical: start with Minimal-MNDM, add complexity only if simpler rhythm and null models fail under held-out prediction and preregistered robustness criteria.

3.7. Neurophenomenology as a Tier-2 extension

NDT is motivated by the possibility that first-person reports of clarity, presence, and temporal organization covary with neural denoising dynamics. However, current experience-sampling methods are generally too slow and noisy to adjudicate sub-second claims about pulsed experience. We therefore

treat neurophenomenological links as Tier-2: useful for future validation, but not part of the Stage-1 evidence required for the core MNDM/RVC/MNPS tests.

The immediate empirical program uses third-person signatures only: event-locked entropy, rhythm proxies, MNPS topology, recurrence, and null-model comparison. Claims about embodiment, presence, insight, depersonalization, or self-prior strength are reserved for appendix-level EAP work and later studies with direct first-person measurement.

3.8. Landscape learning as a follow-on module

The MNDM specifies denoising dynamics given a potential landscape $F(X, t)$ and, in the extended model, an attention geometry $\boldsymbol{G}_t$. How those structures are acquired is an important but separate question. In this core article we therefore treat landscape learning as a **follow-on module**, not as a prerequisite for testing Minimal-MNDM, RVC, or MNPS signatures.

A plausible extension is that wake prediction errors sculpt local basins, NREM replay stabilizes task-relevant regions, and REM-like high-variance periods regularize over-sharp minima. These ideas motivate later experiments on replay, consolidation, and development, but the Tier-1 tests below require only the observable trajectory, rhythm proxy $\hat{r}(t)$, and MNPS features. Failure of a specific wake/NREM/REM learning rule would therefore revise the learning module without falsifying the core rhythmic-geometry claim.

Detailed learning equations, replay kernels, and developmental hypotheses are best handled in dedicated follow-up manuscripts or appendix material once the Stage-1 measurement contract has been validated across public datasets.

The Hippocampal–Entorhinal Landscape System (HELS) is one such prospective contact point: replay and history may update a context-conditioned landscape F_c, with primary future endpoints defined as later transition probabilities, dwell, corridor reachability, or hysteresis rather than a mean shift in (m, d, e). Current HELS work validates analysis infrastructure and eligibility contracts, not a landscape-writing mechanism. It therefore remains distinct from both the online MNDM dynamics and the Stage-1 measurement claim.

4. Rhythmic Variance Control (RVC) and TRN gating

The variance schedule $r(t)$ is not an abstract control parameter but a measurable neural signature: cross-frequency coupling (CFC) in wake and sleep provides candidate timing structure for discrete denoising epochs [15]. The thalamic reticular nucleus (TRN) is a plausible anatomical substrate for state-dependent thalamocortical gating [16], [17], but it is not identified here as the executor of NDT's variance schedule. This functional RVC formulation yields testable predictions: Denoising should occur in discrete "steps" time-locked to specific oscillatory events, not as smooth continuous processes.

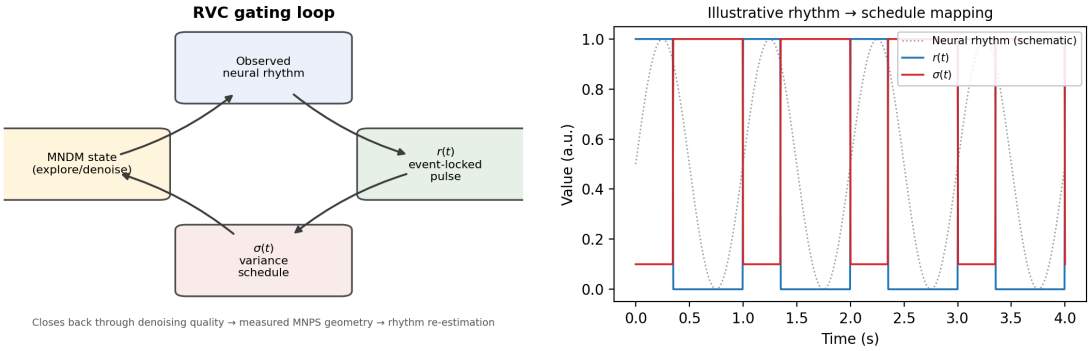


Figure 3: The RVC gating loop, generated by `src/make_figures.py`. An observed neural rhythm is summarized into the event-locked control $r(t)$, which sets the variance schedule $\sigma(t)$ that alternates the MNDM between exploration and denoising; the resulting state and its measured MNPS geometry feed back into how $r(t)$ is estimated in later windows. The right panel shows an illustrative (non-fitted) rhythm-to-schedule mapping for a single oscillatory band; empirical constructions use Eq. 10.

4.1. Biological implementation of r(t)

Distinct implementations across states. The mapping from rhythms to $r(t)$ differs qualitatively between wake and sleep:

- **During wake:** Theta–gamma PAC provides **continuous modulation**—$r(t)$ varies smoothly with task demands (high during focused attention, lower during mind-wandering). Within each theta cycle (~100–200 ms), high-gamma content is gated into and out of local circuits, creating rapid variance oscillations that implement moment-to-moment denoising during perception and action.
- **During NREM:** SO → spindle → ripple nesting creates **discrete, event-locked denoising epochs**. Here $r(t)$ is approximately piecewise constant: $r \approx 1$ during SO–spindle–ripple complexes (strong denoising), $r \approx 0.2$ between complexes (consolidation pause).
- **During REM:** $r(t)$ decreases globally (elevated $\sigma(t)$) while theta coordinates exploratory search, enabling escape from local minima and integration across contexts.

Formal construction from data. For empirical work, $r(t)$ can be constructed as a weighted combination of measurable rhythmic features:

$$r(t) = \mathrm{sigm}\Big(w_\theta \mathrm{PAC}_{\theta\to\gamma}(t) + w_{\mathrm{SO}} A_{\mathrm{SO}}(t) \mathrm{Prec}_{\text{SO-spindle}}(t) + w_\varphi \Phi_{\mathrm{coh}}(t) + b\Big) \tag{10}$$

Important. This parametrization is **not** required by the theory—MNDM only requires that $r(t) \in [0, 1]$ varies with neural rhythms. The specific form above is a **practical implementation** that makes $r(t)$ directly measurable from EEG/MEG/LFP. Alternative constructions (for example, state-specific with separate wake/sleep weights, or purely event-detection-based during sleep) are equally valid and may better capture the continuous vs. discrete distinction noted above.

Section takeaway (RVC for Tier-1 tests). For Tier-1 analyses, RVC enters in two ways: (i) we construct a proxy $\hat{r}(t)$ via Eq. 10 (or closely related variants) to test for event-locked, step-wise entropy changes relative to rhythms; and (ii) we use the same proxy to define schedule null models (for example, smooth vs. event-locked variants) when contrasting Minimal-MNDM against continuous-precision alternatives. Strong evidence for NDT requires that step-like signatures are tied specifically to event-locked portions of $\hat{r}(t)$ rather than to any smooth envelope.

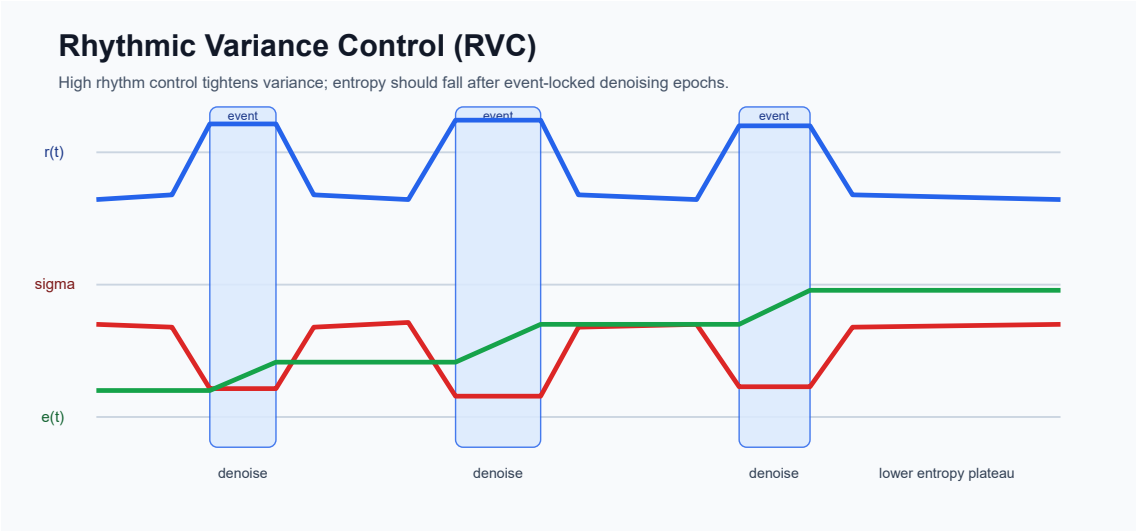


Figure 4: Rhythmic Variance Control. Event-locked increases in rhythmic control $r(t)$ are predicted to lower $\sigma(t)$ anti-phasically and produce step-wise entropy reductions relative to phase-shuffled or smooth-precision null models.

Candidate TRN contribution (summary). Rhythmic proxies $\hat{r}(t)$ derived from PAC, SO–spindle coupling, and phase coherence can be evaluated by convergent and predictive validity without identifying a unique anatomical controller. TRN physiology motivates one candidate substrate, but the dedicated PV/SOM auditory TRN reanalysis recorded in Zenodo 21774408 supports only a bounded PV-linked contraction of MGB population noise dimensionality; it did not validate a general single-unit variance-gating mechanism. Consequently, TRN causality, $r_{\text{true}}(t)$, and a conductance-to-$\sigma(t)$ mapping remain Tier-2 hypotheses. Appendix Section 16.8 records their status and the context-conditional transition-geometry alternative [18].

4.2. Sleep as curriculum and wake as online control

Sleep architecture: Two complementary patterns. Sleep implements denoising through two nested schedules operating at different timescales:

Pattern 1 — Within-cycle dynamics (~90-minute timescale):

- **NREM phase:** High SO–spindle precision ($r(t) \approx 0.8$–0.9) $\Rightarrow$ tight variance windows that **centralize** trajectories in MNPS (manifold contraction toward hubs, consolidation of replayed content).
- **REM phase:** Loose variance with elevated theta ($r(t) \approx 0.2$–0.4) $\Rightarrow$ controlled **exploration** away from local minima (manifold expansion, connector reconfiguration, landscape smoothing).

Pattern 2 — Across-cycle curriculum (full-night timescale):

- **Early cycles (first 2–3):** NREM phases achieve strong centralization (deep hub formation), but trajectories start from **dispersed initial states** post-wake. This creates **large-amplitude oscillations** in F: deep descents during NREM, substantial rebounds during REM as the system explores broadly. **Exploration dominates the cycle envelope**—the landscape is still being surveyed.
- **Late cycles (last 2–3):** NREM phases refine already-centralized topology (hub strengthening), starting from **near-optimal states** left by previous cycles. This creates **small-amplitude oscillations** in F: modest descents during NREM (fine-tuning), limited REM excursions (local refinement). **Integration dominates**—the landscape is nearly settled.

Resolving the apparent paradox. "Early NREM centralizes" and "early cycles explore" are both true but at different levels. **Within** the first cycle, NREM centralizes trajectories; **across** cycles, the first cycle exhibits larger F range (exploration–consolidation amplitude) because it operates on an unrefined landscape. Late cycles show smaller F range because the system is already near target configuration—less distance to consolidate, less benefit to explore.

Quantitative amplitude decay. Let the per-cycle amplitude be

$$\Delta F_n = F_{\text{REM-peak}}(n) - F_{\text{NREM-trough}}(n) \tag{11}$$

where $n = 1, 2, \ldots$ indexes sleep cycles. NDT predicts an exponential envelope:

$$\Delta F_n = \Delta F_0 e^{-\frac{n}{\tau_{\text{curriculum}}}} + \Delta F_{\min}, \quad \tau_{\text{curriculum}} \in [2, 3] \ \text{ cycles (typ.)} \tag{12}$$

Estimation. Compute subject-wise ΔF_n from event-aligned F (or its normalized proxy $\tilde{F}$), fit $(\Delta F_0, \tau_{\text{curriculum}}, \Delta F_{\min})$ via robust nonlinear least squares with bootstrap confidence intervals; control for total sleep time and stage proportions. **Individual differences:** The decay rate $\frac{1}{\tau_{\text{curriculum}}}$ should positively correlate with overnight learning gains (bidirectional prediction: better learning drives efficiency, or efficient consolidation drives learning) and with baseline MNPS component profiles.

Wake. During tasks, theta–gamma PAC is a candidate marker for short denoising epochs aligned to behavioral sampling [19]. Frontopulvinar, mediodorsal-thalamic, and TRN circuits are plausible contributors to state-dependent control [17], but the controller of $r(t)$ is not identified by the present framework.

4.3. Geometric and behavioral signatures

RVC predicts specific, testable mappings between rhythms and MNPS geometry:

- **Local curvature smoothing:** Increases in theta–gamma PAC strength/consistency $\Rightarrow$ reduced local curvature and more stable geodesics along task-aligned trajectories (inverted-U with performance under load).
- **Hub accessibility:** Higher SO–spindle phase precision $\Rightarrow$ greater access to consolidation hubs (higher hub betweenness/centrality) and steeper overnight entropy drops in relevant subnetworks.
- **Connectivity and flow:** Stronger cross-regional phase coherence $\Rightarrow$ improved manifold connectivity (fewer fragmented components) and more reliable transition connectors.

4.4. Measurement and perturbation plan

Measurement strategy. We estimate $r(t)$ from EEG/MEG/LFP using validated methods: (i) phase–amplitude coupling via modulation index or mean vector length (Tort et al., 2010; see also [15], [20]); (ii) SO/spindle detection via amplitude thresholding or wavelet transforms, with phase precision quantified by circular statistics (for example, Mölle et al., 2011); (iii) cross-regional phase coherence via weighted phase lag index (wPLI) or phase-locking value (PLV); and (iv) coupling within SO–spindle–ripple coordination in intracranial and scalp data [21], [22], [23].

Operational definitions and multiple-comparisons control. For event-locked analyses we use pre-registered windows (baseline: $[-1.0, 0]$ s; post-event: $[0, 1.5]$ s relative to SO–spindle onset), sliding-window entropy (500 ms, 50 % overlap), and alignment latencies constrained to biologically plausible ranges (spindle onset within ±200 ms of SO up-state). Detection thresholds follow literature priors (SO: 0.5–1 Hz band, amplitude >75th percentile; spindles: 11–16 Hz, duration 0.5–2 s). Family-wise error is controlled with Benjamini–Hochberg FDR across channels and parameter multiverse. Effect-size priors from recent syntheses (for example, [24]) set expectations of $\Delta e \approx 15$–30 % and medium effects ($d \sim 0.4$–0.6) for coupled events.

For step-wise entropy analysis: Compute spectral/sample entropy in sliding windows (for example, 500 ms), detect SO–spindle or theta peaks, align entropy time series to these events, and test for systematic **drops** time-locked to events (permutation testing against phase-shuffled controls). **Key prediction:** Entropy should drop **after** event onset with timing that scales with event duration (longer spindles $\rightarrow$ deeper/longer entropy reduction).

Perturbation strategy. Closed-loop stimulation provides causal tests: (i) tACS at theta or SO frequencies can enhance or disrupt natural coupling; (ii) stimulation at anti-phase (for example, 180° relative to detected SO) should specifically disrupt **temporal ordering** of denoising steps (NDT predicts items encoded during anti-phase show false temporal associations), while in-phase stimulation

enhances them; (iii) in animals, optogenetic manipulation of TRN sectors tests whether variance control is necessary and sufficient for the predicted geometric changes.

Failure modes and perturbation roadmap. Schedule dissonance—weak, desynchronized, or pathologically rigid rhythm control—should alter recurrence, connectors, and manifold accessibility. Pharmacological and stimulation studies may eventually test this causal chain, but such intervention claims are Tier-2 and require dedicated safety, dosing, and preregistration. In this core article they serve only as a roadmap for future perturbation work, with readouts at the level of component readouts rather than a single health score.

Falsifiable contrasts. If RVC genuinely implements **discrete** steps, entropy declines should be step-wise and event-locked; anti-phase disruption should scramble **order** (temporal misbinding), not just reduce overall precision. Smooth, monotonic entropy changes and insensitivity to timing would falsify this mechanism.

Embodied Anchoring Principle (EAP; Tier-2 extension). EAP treats denoising as conditioned on interoceptive state via a self-prior and body-synchronized variance gating. The full formalization, interoceptive pipeline, and phenomenology (including equations and operationalization details) are moved to Appendix Section 16.7.

5. Measurement layer: MNPS

The MNDM specifies dynamics in an abstract noetic state space X. To make these dynamics empirically observable, we construct the **Meta–Noetic Phase Space (MNPS)**—a measurable coordinate system derived from neural recordings. The Noetic Atlas readouts built on top of MNPS are described in Appendix Section 16.2 so the main text can focus on the canonical Stage-1 trajectory and rhythm tests.

5.1. From neural data to MNPS coordinates

We extract three families of features from EEG/MEG/fMRI time series, corresponding to the core MNPS axes (m, d, e) and MNDM terms:

Entropy/Disorder (e): Quantifies local uncertainty and signal complexity.

- **EEG/MEG:** Spectral entropy (Shannon entropy of power spectrum, 1–40 Hz), sample entropy, Lempel–Ziv complexity, multiscale entropy.
- **fMRI:** BOLD signal variance per parcel, sample entropy of BOLD time courses, regional activation dispersion.

Mobility / rhythmic coherence (m): Quantifies structured oscillatory organization and coordination.

- **EEG/MEG:** Phase–amplitude coupling (theta–gamma, alpha–gamma), phase-locking value (PLV) or weighted PLI (wPLI) across regions, spectral power in canonical bands (theta, alpha, beta, gamma).
- **fMRI:** Ultra-slow oscillation amplitude (0.01–0.1 Hz), quasi-periodic pattern (QPP) strength, BOLD–respiratory coupling.

Diffusivity / deviation from metastability (d): Quantifies distance from optimal integration–segregation balance.

- **EEG/MEG:** Participation ratio (inverse concentration of power across channels/components), graph modularity Q.
- **fMRI:** Integration index (for example, $|Q - Q^*|$ where $Q^* \approx 0.3$–0.5 is near-optimal), global efficiency, synchrony variance.

These features are computed in sliding temporal windows (for example, 2 s for EEG, TR-matched for fMRI) and z-scored within subject to yield comparable (m, d, e) time series that define MNPS coordinates.

Note on spatial scope (global vs. regional). In this first NDT manuscript we primarily treat X_t as a **global** noetic state, so m aggregates rhythmic structure across the brain (for example, mean PLV/ wPLI across channel pairs, and pooled band-power indices). This inevitably mixes spatial scales but keeps the minimal MNDM tractable; regional heterogeneity is addressed in separate work on Regional MNPS, where each network has its own local (m, d, e) chart and block-Jacobian. When we use pairwise measures such as PLV in the present 3D MNPS, they are explicitly summarized (for example, robust mean across pairs) rather than attributed to single regions, and the resulting global X_t should be read as a coarse, whole-brain state variable.

Definition vs. implementation. At the theory level, e, m, and d denote **classes** of entropic, mobility/rhythmic-coherence, and metastability-deviation measures: any choice within these classes that respects basic regularity assumptions defines a valid MNPS chart. In this manuscript, we instantiate e, m, and d with spectral entropy, composite PAC/PLV/power indices, and modularity-based deviation, but our primary predictions target properties that should be stable across this class: topological organization (component fragmentation, hub structure), the timing of event-locked entropy changes relative to rhythms, and the qualitative structure of flows in MNPS. Chart-dependent quantities (exact Δe magnitudes in a given dataset, or absolute composite scores) are always reported together with the specific implementation choices used to compute them.

An extended, 9-dimensional **Stratified MNPS** framework that decomposes the core (m, d, e) axes into sub-coordinates (for example, attentional vs. limbic mobility, local vs. network diffusivity) is developed in a separate working paper [25]. The present manuscript restricts itself to the 3D MNPS needed for the core NDT predictions, but all constructions here are designed to remain compatible with that stratified extension.

Measurement contract and later extensions. For this article, the canonical object is the 3D trajectory $\boldsymbol{X}_{\mathrm{MNPS}}(t) = [e(t), m(t), d(t)]^T$ plus its rhythm proxy $\hat{r}(t)$. Atlas component readouts are optional appendix summaries. Stratified 9D MNPS, Meta-Noetic Jacobians, reachability cones, regional MNPS, and cytoskeletal or dendritic accessibility models are later extensions. They may refine mechanism and regime interpretation, but they should not be treated as assumptions needed to evaluate the Stage-1 MNDM/RVC/MNPS predictions.

5.1.1. Algorithmic specification for MNPS construction

To ensure reproducibility, we provide explicit computational steps for constructing MNPS coordinates from neural data.

Step 1: Preprocessing.

1. **EEG/MEG:** Band-pass filter 0.5–100 Hz, notch-filter 50/60 Hz, artifact rejection (ICA or robust averaging), re-reference to average.
2. **fMRI:** Standard preprocessing (motion correction, spatial normalization, parcellation via Glasser-360 or Schaefer-400), temporal filtering 0.01–0.1 Hz.

Step 2: Feature extraction with explicit formulas (EEG/MEG).

- **Entropy $e(t)$:**

$$e_{\mathrm{spectral}}(t) = -\sum_{f=1}^{40} P_{f(t)} \log P_{f(t)}, \quad P_{f(t)} = \frac{S_{f(t)}}{\sum_{f'} S_{f'}(t)} \tag{13}$$

  where $S_{f(t)}$ is power spectral density at frequency f in a window centered at t (estimated via Welch's method; window 2 s, 50 % overlap).

- **Rhythmic coherence $m(t)$:**

$$m(t) = w_{\mathrm{PAC}} \mathrm{PAC}_{\theta\to\gamma}(t) + w_{\mathrm{PLV}} \overline{\mathrm{PLV}}_{\theta}(t) + w_{\mathrm{pow}} P_{\mathrm{theta}}(t) \tag{14}$$

  where PAC is a modulation index [20], $\overline{\mathrm{PLV}}$ the mean phase-locking value across electrode pairs, and P_{theta} theta-band power; default weights might be $w_{\mathrm{PAC}} = 0.5$, $w_{\mathrm{PLV}} = 0.3$, $w_{\mathrm{pow}} = 0.2$.

- **Deviation from metastability $d(t)$:**

$$d(t) = \mid Q(t) - Q^* \mid, \quad Q^* \approx 0.4 \tag{15}$$

where $Q(t)$ is graph modularity at time t, computed on a thresholded phase-coherence matrix (for example, top 10 % of connections).

Step 3: Normalization and MNPS assembly.

1. z-score each feature within subject: $e_{\text{norm}}(t) = \frac{e(t)-\mu_e}{\sigma_e}$ etc.
2. Construct coordinate vector $\boldsymbol{X}_{\text{MNPS}}(t) = [e_{\text{norm}}(t), m_{\text{norm}}(t), d_{\text{norm}}(t)]^T$.
3. Optionally expand with higher-order features (for example, band-specific PAC, regional entropy) for a higher-dimensional MNPS.

Pseudocode.

1. **Preprocess.** Filter and clean the neural data according to `params.filter_specs` to obtain `data_clean`.
2. **Windowing.** Extract overlapping windows of length `params.window_s` with 50 % overlap, yielding `windows`.
3. **Feature extraction.** For each window:
   - Compute spectral entropy $e[t]$ from the Welch PSD.
   - Estimate theta–gamma coupling (`pac`), mean PLV (`plv`), and theta-band power (`theta_pow`) to form $m[t]$.
   - Build the phase-coherence matrix, compute modularity Q, and derive $d[t] = \mid Q - 0.4 \mid$.
4. **Normalization.** e, m, and d are z-scored within subject to yield $e_{\texttt{norm}}$, $m_{\texttt{norm}}$, and $d_{\texttt{norm}}$.
5. **Return.** The MNPS coordinates are $\text{MNPS}_{\text{coords}}(t) = [e_{\texttt{norm}}(t), m_{\texttt{norm}}(t), d_{\texttt{norm}}(t)]^T$.

Coordinate invariance and robustness. While specific choices for e, m, and d define a particular MNPS chart, we explicitly define **invariance classes**— predictions that must hold across all valid coordinate charts to be considered robust:

1. **Topological Invariance:** Qualitative features like component fragmentation (in psychosis) or collapsed manifold volume (in depression) must remain statistically significant regardless of whether e is Spectral or Sample entropy.
2. **Temporal Invariance:** The **timing** of step-wise entropy drops relative to spindles must be preserved across feature choices, even if absolute magnitudes shift.
3. **Gradient Invariance:** The direction of flow ($d\frac{X}{d}t$) in local regions should correlate ($r > 0.7$) across chart choices after Procrustes alignment.

Primary claims target these intrinsic properties; chart-level metrics are used comparatively within fixed pipelines.

Appendix pointer. The Noetic Atlas applies embedding, topology, recurrence, and metastability tools to MNPS trajectories; those component definitions, along with the optional Stage-1 NDHI composite, are provided in Appendix Section 16.2. The core predictions below depend on the MNPS trajectory, rhythm proxy, and prespecified component-level contrasts rather than on the Atlas as an additional theory layer.

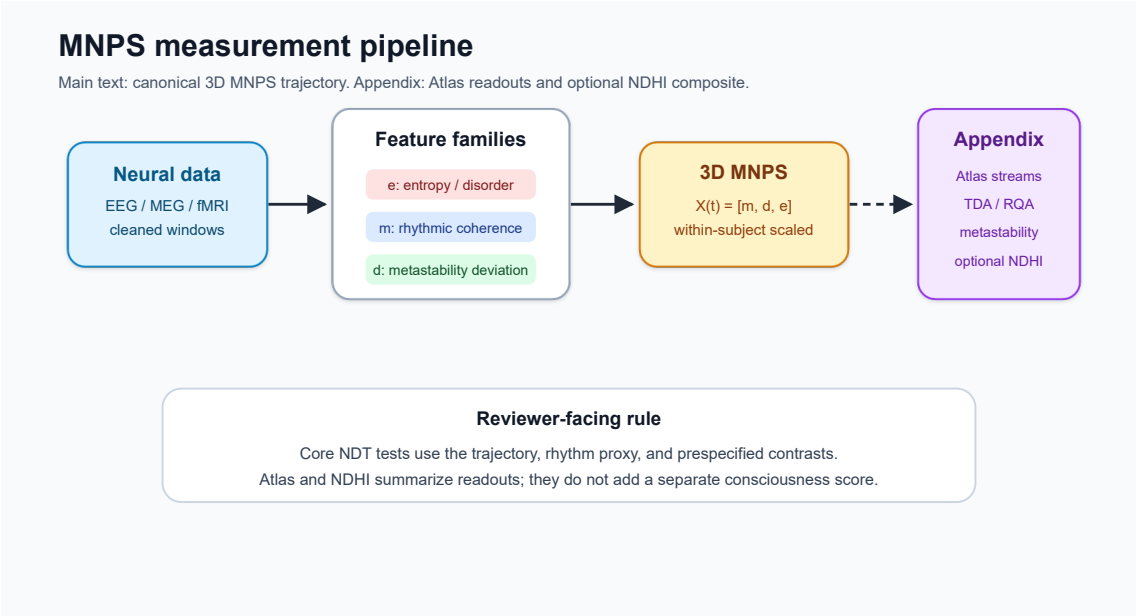


Figure 5: MNPS measurement pipeline. Neural recordings are windowed into the three Stage-1 coordinate families (m, d, e); appendix Atlas tools and NDHI are downstream summaries rather than additional assumptions of the core model.

6. Predictions and falsifiable claims

NDT makes specific, testable predictions that distinguish it from alternative frameworks. We organize these by type: **signature predictions** (novel patterns NDT uniquely predicts), **competitive predictions** (where NDT and alternatives make different predictions), and **falsification criteria** (observations that would disprove core claims).

For empirical planning we distinguish a small **Tier-1 primary test set** from longer-horizon or more speculative **Tier-2** predictions. The Tier-1 set comprises: (P1) Signature Prediction 1 (step-wise NREM entropy drops time-locked to SO–spindle events), (P2) Signature Prediction 3 (sleep curriculum with NREM centralization and REM expansion in MNPS topology), and (P3) bounded clinical geometry contrasts such as collapsed, over-recurrent trajectories versus fragmented, shallow-recurrence trajectories. EAP-specific, neurophenomenological, therapy, and detailed disorder-subtype predictions are Tier-2: they guide future work but are **not** required to succeed in the first wave of empirical tests for NDT to be considered supported.

Subsequent empirical evaluations. Later NDT manuscripts report provisional support for traversability dissociation, MNJ flattening under propofol, regional MNPS signatures in fMRI, and extreme-regime dissociations. These results are treated as follow-on empirical tests that inform effect-size priors and model refinement [26], [27], [28], [29].

A paired hypoglycemia–euglycemia EEG reanalysis adds a complementary boundary: it reports a reproducible stratified MNPS condition shift but no FDR-supported reachability collapse. That study contains no continuous glucose, autonomic, interoceptive, recovery, or behavioural measurement and therefore does not test an EAP metabolic gate or a consciousness readout.

Current boundary from later work. Subsequent analyses sharpen an important interpretive constraint: reduced dynamical capacity should not be equated with unconsciousness. Physiological sleep, pharmacological suppression, and seizure states can all reduce accessible dynamics, but they may differ in transport-innovation balance and pathological lock-in. The core NDT article therefore avoids a single scalar "consciousness level" readout. When later work uses reachability, traversability, or Meta-Noetic Jacobians, those measures are treated as second-order extensions that decompose

capacity, transport capture, and canalization rather than replacing the Stage-1 MNDM/RVC/MNPS tests.

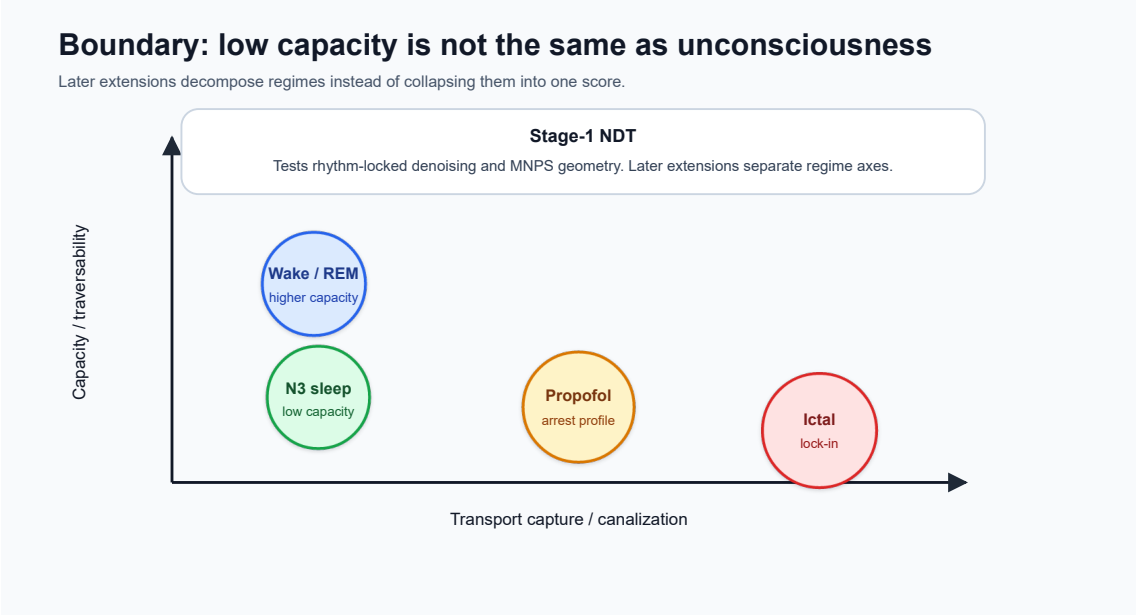


Figure 6: Boundary inherited from later NDT work. Reduced capacity or traversability should not be collapsed into a single consciousness score; later extensions separate capacity, transport capture, and pathological lock-in.

Effect size priors and thresholds. Where we quote specific effect sizes (for example, $\Delta e \approx 20$–$35\,\%$ or $d > 0.6$), these are grounded in a combination of (i) literature-based priors—especially recent syntheses on SO–spindle mechanisms and sleep-dependent entropy changes (for example, [24])—and (ii) simulation results using the minimal MNDM that indicate the scale at which predicted topology/entropy signatures become reliably detectable under realistic noise and sample sizes. We therefore treat thresholds such as $d \geq 0.6$ or $r \geq 0.3$ as conservative targets for **strong support**, while smaller but consistent effects are still informative as **moderate** or **exploratory** evidence, provided they meet the preregistered robustness criteria described in the methods section.

Overview. The five predictions below are summarized here for quick reference; each is unpacked in its own subsection.

#	Tier	Prediction	Primary test
1	Tier 1	Step-wise entropy locked to rhythmic events	NREM entropy drops time-locked to SO–spindle events, not smooth decay.
2	Tier 2	Phase-specific denoising during wake	Theta-phase-locked modulation of recall accuracy and local entropy.
3	Tier 1	Sleep as geometric curriculum	NREM centralization vs. REM expansion in MNPS topology across the night.
4	Tier 1	Bounded clinical geometry contrasts	Reproducible manifold deformation across a clinical contrast exists (observed: medication-stratified FEP m/path-cost shift; BDI-linked depression reachability and 9D MNJ); original Mapper/DET and volume-collapse operationalizations remain open, untested-as-stated Tier-2 targets.

#	Tier	Prediction	Primary test
5	Tier 2	Anti-phase disruption affects temporal order (competitive prediction)	Order errors are disproportionately elevated relative to item-memory errors under anti-phase rhythmic stimulation.

Overview of NDT's signature and competitive predictions, with tier and primary test. See the falsification-criteria table below (Section 6.6) for the corresponding falsifiers.

6.1. Signature Prediction 1: Step-wise entropy locked to rhythmic events

Claim. During NREM sleep, neural entropy should decline in **discrete steps** time-locked to SO–spindle (and ripple) events, not smoothly.

Operationalization. Detect SO–spindle complexes in sleep EEG; compute spectral/sample entropy in 500 ms windows; align entropy time series to event onsets; test for systematic drops within 100–1000 ms post-onset.

Predicted pattern. Entropy drops $\Delta e \approx 15$–$30\,\%$ within one spindle duration (~0.5–1.5 s) after complex onset, with effect size $d > 0.6$ vs. pre-event baseline. Longer/higher-amplitude spindles → deeper drops; non-nested spindles or isolated SOs show weaker effects.

Consequence. SO–spindle precision (phase consistency) should predict: (i) magnitude of overnight entropy reduction ($r \sim 0.3$–0.5), (ii) next-day memory performance ($r \sim 0.2$–0.4), and (iii) component-level MNPS changes from pre-sleep to post-sleep.

6.2. Signature Prediction 2: Phase-specific denoising during wake

Claim. Items encoded at different theta phases undergo denoising at different depths, affecting encoding quality [30].

Operationalization. Present stimuli at controlled theta phases; measure encoding success and neural entropy as a function of presentation phase.

Predicted pattern. Items near theta peak show better recall and lower entropy. Theta–gamma PAC strength shows an inverted-U with task performance (optimal at intermediate coupling; too weak → insufficient denoising, too strong → over-rigid).

6.3. Signature Prediction 3: Sleep as geometric curriculum

Claim. Manifold topology transforms systematically: NREM centralization, REM expansion, progressive overnight refinement.

Predicted patterns:

- **NREM:** Hub betweenness ↑ ($d > 0.5$), manifold volume ↓ (20–40 %), recurrence rate ↑.
- **REM:** Opposite pattern (expansion, decreased centrality).
- **Overnight:** Net centralization with oscillations; late > early NREM in hub coherence.
- **TMR:** Cued content shows increased local Mapper density and steeper local entropy drops.

6.4. Signature Prediction 4: Bounded clinical geometry contrasts

Claim. Some clinical contrasts should produce reproducible manifold deformations, but these are bounded dataset-level hypotheses rather than diagnostic categories.

Observed signatures (internal, dataset-bounded). Two companion analyses report FDR-surviving manifold deformations that support the **existence** half of this claim, though not in the form originally sketched below:

- **First-episode psychosis (medication-stratified):** Across two independent resting-state cohorts, shared-reference MNPS geometry separates FEP from controls, but as a medication-dependent double dissociation rather than a single universal signature: medicated FEP shows a static shift in the mobility/coherence coordinate (m; $d = 0.881$, $q = 0.016$, cohort ds003944), while unmedicated

FEP instead shows elevated trajectory cost ($d \approx 0.77$, FDR-borderline, cohort ds003947)—a static-geometry signature under medication versus a path-cost signature without it.

- **Depression (symptom burden):** In a resting-state cohort split by BDI symptom burden ($n = 121$), the clearest FDR-surviving deformations are in the reachability coordinates (log-determinant and eigenvalue-variance, $|d| \approx 0.68$–0.74) and 9D Meta-Noetic Jacobian rotational power ($d = 0.569$), not in coarse 3D volume.

These results confirm that reproducible, bounded manifold deformation exists across a clinical contrast, while falsifying the specific illustrative operationalizations first proposed here (Mapper fragmentation for psychosis; 3D volume collapse for depression), which remain open, untested-as-stated Tier-2 targets:

- **Depression (original target):** Collapsed volume ($d > 0.6$ vs. controls), over-recurrence, flattened curvature, reduced hub access [31].
- **Psychosis (original target):** Fragmented Mapper (high component count), shallow recurrence (low DET), unstable geodesics [32].

Negative and null results bound both signatures further: psychosis turning-angle/geodesic disruption is FDR-null under a medication-stratified design, and depression's paired task–rest deltas are broadly null despite state-level signal. The full evidence ledger—cohorts, effect sizes, and all negative results—is maintained in `articles/NDT/parts/008_20260809_clinical_disorder_evidence_survey.md`.

PTSD, anxiety, neurodegeneration, and anesthesia are not primary claims of this paper. They are better handled as follow-on regime tests where capacity, traversability, transport capture, and pathological lock-in can be analyzed separately; companion Parkinson and Alzheimer/FTD analyses are cataloged in the same evidence ledger as Tier-2 material.

6.5. Competitive Prediction 5: Anti-phase disruption affects temporal order

NDT prediction. Anti-phase rhythmic stimulation disrupts **temporal ordering** of denoising steps, leading to temporal binding errors and false associations.

Predictive-coding prediction. Disruption reduces overall precision, leading to proportional impairment of all memory.

Distinguisher. Order errors become disproportionately elevated vs. item memory under anti-phase stimulation.

6.6. Falsification criteria

NDT would be falsified or require major revision if:

#	Observation	Implication
1	Entropy declines are smooth/continuous, not event-locked.	Contradicts rhythmic variance control.
2	Manifold geometry is invariant across wake/sleep/pathology.	Contradicts geometry–dynamics coupling.
3	Anti-phase disruption affects only magnitude, not temporal structure.	Contradicts discrete-step denoising.
4	All effects are derivable from precision-only models without discrete steps or geometry.	NDT adds nothing beyond existing precision-scheduling accounts.
5	Tier-2 phenomenological variables remain unrelated to (m, d, e) or Ω under adequate temporal alignment.	Constrains the phenomenological extension only; does not by itself falsify the Stage-1 dynamical model.

Falsification criteria for NDT's Stage-1 and Tier-2 claims.

6.7. Failure modes, confounds, and unfalsifiability patterns

We enumerate concrete failure modes and how the preregistered analysis circumvents unfalsifiability:

- **Chart-flexibility inflation:** Post-hoc switching of MNPS features (m, d, e) could manufacture effects. **Guardrail:** Predefine a multiverse grid that explicitly includes **mechanistically distinct** feature implementations (for example, spectral vs. Lempel–Ziv entropy, PAC-heavy vs. PLV-heavy m) rather than only parameter tweaks of a single measure, and require directional consistency across ≥80 % of specifications with FDR control; report Procrustes alignment between MNPS variants (target $R^2 > 0.7$). Core topology (for example, depression-like manifold collapse) must survive across these distinct implementations to count as robust.
- **Schedule over-flexibility:** Allowing $r(t)$ to be freely re-fit risks tautology. **Guardrail:** Construct $\hat{r}(t)$ from observables with fixed methods and cross-validated weights; treat $\sigma(t)$ amplitude as the only free scalar in Stage 1 estimation.
- **Event-detection circularity:** Using entropy changes to define events would guarantee step-like drops. **Guardrail:** Detect SO/spindles with canonical band-limited criteria and circular statistics independent of entropy, then test entropy time-locked to those events.
- **Parameter overfitting:** Rich parameterizations (for example, adaptive $\boldsymbol{G}_t$) may fit noise. **Guardrail:** Nested model comparison with BIC and held-out prediction; accept added complexity only with Δ BIC > 10 and >10 % test-set error reduction.
- **Hidden confounds:** Vigilance, motion, and medication can mimic topology changes. **Guardrail:** Regress standardized confounds, analyze within-subject stage contrasts, and verify absence of effects in phase-randomized surrogates.
- **Unfalsifiability pattern A (moving targets):** Redefining predictions after seeing data. **Guardrail:** Preregister operationalizations, parameter grids, and success thresholds.
- **Unfalsifiability pattern B (equivalence classes):** Claiming success under transformations that preserve ordering. **Guardrail:** Use interventions that break equivalences (anti-phase rhythmic stimulation; TRN-targeted pharmacology) and require order-specific effects.

In addition, we include **null-model controls** (random-walk trajectories and continuous-precision models) alongside Minimal-MNDM in simulations and empirical fits; step-wise, event-locked entropy drops and rhythm-locked topology changes must exceed these nulls under identical preprocessing and FDR control.

7. Simulated demonstrations

To demonstrate that the MNDM can reproduce key theoretical predictions, we simulate the minimal formulation (Eq. 1) under several scenarios capturing core dynamics: wake-to-sleep transitions, sleep cycling with curriculum structure, and disorder-typical perturbations. All simulations use Euler–Maruyama integration; predictor–corrector samplers from score-based diffusion can be used analogously for improved stability [33].

Minimal worked example and null models. Before complex scenarios, we present a worked example contrasting Minimal-MNDM against two nulls:

- **Minimal-MNDM (event-locked variance):** $dX_t = -\nabla F dt + \sigma(t) dW_t$, $F = E - R$, with $\sigma(t) = \sigma_{\min} + \sigma_0(1 - r(t))$ and piecewise-constant $r(t)$ (1.0 during detected events, 0.2 otherwise). This square-wave idealization is best read as an analytically convenient limit case corresponding to strongly pulsatile gating; in later scenarios we also consider smooth rhythm proxies passed through the non-linear construction in Eq. 10 to assess whether step-like entropy dynamics can **emerge** from graded inputs rather than being assumed.

- **Null A (random walk in MNPS):** $dX_t = \sigma_{\mathrm{RW}} dW_t$ with identical preprocessing, feature extraction, and windowing; no coupling to $r(t)$.
- **Null B (continuous-precision PC-like):** $dX_t = -\nabla F dt + \sigma_{c(t)} dW_t$ with smooth $\sigma_{c(t)}$ obtained by low-pass filtering $\hat{r}(t)$ (no event steps).

Readouts: (i) event-triggered spectral entropy $e(t)$ (500 ms windows), (ii) Mapper fragmentation index, (iii) RQA determinism (DET). **Expected pattern:** Minimal-MNDM shows step-wise post-event Δe and rhythm-locked topology changes; Null A shows no event-locked structure; Null B shows smooth entropy drifts without discrete steps.

7.1. Scenario 1: Wake-to-sleep transition

Setup. We simulate a single trajectory over $T = 3600$ s (1 h) with $r(t)$ transitioning from wake-like continuous modulation to sleep-like discrete epochs.

Parameters (illustrative):

- $X \in \mathbb{R}^{50}$ (low-dimensional noetic manifold).
- $e(X) = \left(\frac{1}{2}\right) \mid X \mid^2$ (quadratic entropy surrogate).
- $m(X) = \sum_{i<j} \cos\left(X_i - X_j\right)$ (pairwise coherence).
- $\sigma_0 \approx 0.5$ (chosen to match empirical EEG-feature variability).
- $r(t)$ decreases from 0.4 (wake) to 0.9 (deep sleep) over the hour.

Results. Trajectories show (i) progressive descent in $F(X, t)$ as variance decreases, (ii) manifold-volume contraction (lower participation ratio), and (iii) increased recurrence to low-F basins. Entropy $e(X)$ drops from ~15 (wake-like) to ~5 (sleep-like), while coherence $m(X)$ increases from ~0.3 to ~0.8, consistent with "centralization" during NREM.

7.2. Scenario 2: NREM/REM cycling

Setup. We simulate an 8-hour sleep night with four 90-min cycles, each containing NREM (tight variance) and REM (loose variance) phases.

Parameters (schematic): $r_{\mathrm{NREM}} \approx 0.85$, $r_{\mathrm{REM}} \approx 0.3$, with gradual ramps; NREM duration decreases across cycles, REM increases.

Results. Trajectories exhibit the dual-timescale curriculum architecture:

- **Within-cycle pattern:** Manifold contracts during NREM, expands during REM, visible as oscillations in participation ratio and geodesic distances.
- **Across-cycle pattern:** Early cycles show large-amplitude F oscillations (exploration-dominated), late cycles small-amplitude oscillations (integration-dominated). Final REM ends at much lower F than initial wake, confirming net overnight consolidation despite REM-driven exploration.
- **Entropy dynamics:** Step-wise drops during NREM epochs (20–30 % per spindle-dense period), partial rebounds in REM (10–15 %), with overall downward trend—matching the dual-schedule architecture.

7.3. Scenario 3: Bounded parameter perturbations

Starting from a reference baseline, we perturb parameters to emulate two broad geometry regimes:

- **Depression simulation:** Flatten ∇F (weaker score) and reduce exploratory variance (σ_0 small) with normal $r(t)$. Result: Trajectory becomes trapped in a high-F local minimum; manifold shows collapsed volume, excessive recurrence, and flattened curvature— matching predicted depression topology.
- **Psychosis simulation:** Elevate noise (σ_0 large), weaken gradient, and desynchronize $r(t)$ by adding noise. Result: Trajectory wanders chaotically, rarely settling into coherent basins; manifold shows fragmented structure, shallow recurrence, and unstable geodesics.

7.4. Scenario 4: Learning-module placeholder

Learning-induced changes in F and G_t are not simulated as a primary claim in this core article. They are reserved for follow-on work once the Stage-1 trajectory, rhythm, and component readouts have been validated.

8. Methods and analysis plan (public datasets)

We outline a preregistered analysis plan for testing NDT predictions on public datasets. By specifying methods, parameters, and success criteria in advance, we aim for transparent, hypothesis-driven evaluation. For fMRI, we focus on topology/dynamics (centralization, fragmentation, hubs) rather than sub-second step-wise effects; EEG/MEG datasets test the temporal predictions.

8.1. Target datasets

Dataset	Modality	What it tests
HCP	Resting-state and task fMRI	Manifold topology predictions and exploratory composite stability in healthy cognition (e.g., component profiles vs. fluid intelligence/working memory; task-dependent centralization vs. expansion).
Sleep-EDF	Polysomnography	Step-wise entropy and sleep-curriculum predictions: NREM centralization, REM expansion, SO–spindle precision vs. entropy drops and next-day performance.
OpenNeuro clinical datasets	EEG/fMRI (e.g., depression, schizophrenia)	Disorder-specific topology predictions (collapsed volume and over-recurrence in depression; fragmented Mapper and shallow recurrence in psychosis, etc.).

Public datasets targeted for Stage-1 and exploratory NDT tests.

Subsequent subsections adapt the MNPS pipeline and Atlas summaries (S1) to these datasets (feature extraction, Mapper/RQA settings, component summaries) and specify effect-size thresholds, power analyses, and multiverse-analysis strategies much as in the LaTeX version.

9. Limitations and future work

We explicitly acknowledge key limitations:

- **Learning problem:** MNDM specifies dynamics given F and G_t but does not yet derive how they are learned from synaptic/circuit plasticity. A simulator-first candidate mechanism—replay-dependent slow branch-level accessibility—reproduces a joint linking/perturbation/rescue signature not matched by nine tested simpler comparators [34], and a follow-on truth-known observability audit separately maps which measurement objects can and cannot recover that mechanism under partial, non-oracle observation (Appendix Section 16.4) [35]. Both remain synthetic model-discrimination and measurement-recoverability results, not validated biological plasticity or a fitted NDT learning rule; the tri-phase learning architecture and HELS remain the open bridge to direct, content-specific transition or reachability tests on real circuits. Existing HELS analyses validate infrastructure, not a mechanism.
- **Measurement and identifiability:** Coordinate choices for MNPS introduce ambiguity; robust multiverse analysis and targeted perturbations are needed to disambiguate models (see below).
- **Phenomenological methods:** Claims about pulsed experience, presence, and embodiment require high-resolution neurophenomenology that is still under development. Operational content sections and fibre language (Appendix Section 16.6) organize hypotheses, and a dedicated measurement-

precursor program has now started: in mouse visual cortex, population decoders trained only on real, orientation-matched edges generalize to illusory-contour trials beyond shuffled-label and region-specificity controls. This is a genuinely blind, frozen content-generalization decoding result, not a phenomenal-report measurement; it neither measures qualia nor closes the hard problem.

- **Clinical translation:** Geometric pharmacology and intervention design are explicitly future-work goals; any clinical pathway claims are provisional until replicated with preregistered, dataset-specific validation.
- **Hard-problem scope:** NDT addresses **when/how** denoising unfolds and what geometry it leaves, not why denoising is accompanied by subjective experience. This is a permanent scope boundary: no measurement-recoverability audit or content-decoding result above is intended or able to close it.

9.1. Identifiability of landscape, geometry, and schedule

At the level of observables, $(F, \boldsymbol{G}_t, r(t))$ are partially non-identifiable: different combinations of potential shape, attention geometry, and variance schedule can yield similar trajectories in MNPS. Our strategy is threefold. First, chart-level ambiguity is addressed by the multiverse guardrails above, which require that qualitative patterns (for example, step-wise NREM entropy drops, depression-like manifold collapse) be consistent across many reasonable (m, d, e) definitions. Second, we rely on **targeted perturbations** that preferentially modulate one component: TRN-focused manipulations can test candidate circuit contributions but cannot be presumed to identify $\sigma(t)$; attentional tasks and fronto-parietal lesions should primarily affect $\boldsymbol{G}_t$; and interventions on plasticity or replay should reshape F itself. Third, we explicitly compare Minimal-MNDM against null models (for example, continuous-precision "Null B") in both simulations and data: patterns such as discrete, event-locked entropy steps or tightly coupled topology changes must be reproducible by Minimal-MNDM but **not** by plausible precision-only models if they are to count as evidence specifically for rhythmic, geometry-coupled denoising.

Future work should unify score-learning with predictive-error plasticity, incorporate action and active inference, model development and lifespan changes in schedules, extend to multi-agent settings, and explore laminar implementations in detail.

10. Conclusion

We summarize the thesis that conscious-state correlates may be studied as **rhythmic, step-wise denoising on a learned manifold**. The core advances are discrete timing via RVC, a minimal MNDM that can be compared against nulls, and geometry-first visibility through MNPS and the appendix Noetic Atlas. NDT is offered as a staged research program with concrete next steps: public-dataset analyses, stricter multiverse controls, and later perturbation experiments once the Stage-1 measurement layer is stable.

11. Acknowledgments

This work emerged through structured correspondence among human and AI collaborators; large language models (GPT-5, Claude, Gemini, Grok) assisted in literature synthesis, drafting, and editorial refinement.

12. Contact

Robin Langell — hello(at)noeticdiffusion.com

13. Licensing

This manuscript provides only a brief licensing summary; full terms and conditions are available via the Mychainos multi-layer licensing framework (`https://mychainos.org/licence/`). In short, the conceptual content is released under CC BY-SA 4.0, software under GPLv3, and relevant hardware/biological artifacts under open hardware/biological licences (for example, CERN OHL v2 and Open-MTA). Unified ethical guardrails apply: no coercive, military, or surveillance applications; no patenting or black-boxing of core methods; and derived works should remain accessible, attributed, and shared with care.

14. Data and Code Availability

Reference implementations and analysis pipelines will be released at `https://github.com/ruppi86/NoeticDiffusion` (public upon acceptance), with licensing following the Mychainos policy.

15. Trademark Notice

Certain terms used in this manuscript (including **Noetic Diffusion**, **Noetic Diffusion Theory**, **Noetic Diffusion Mapping**, **Noetic Diffusion Health Index**, and **Noetic Atlas**) are used as project names and are the subject of ongoing trademark applications. A concise description of the stewardship rationale, current registration status, and how this interacts with open scientific use is available at `http://noeticdiffusion.com/license.html`. These trademark aspects do not affect the scientific content, reproducibility, or licensing of the methods described here.

16. Appendix

The formal core above (Minimal-MNDM, functional RVC, MNPS) is self-contained for Stage-1 testing. The following Tier-2 extensions, methodological audits, and comparative framing are developed in full in dedicated supplements (**Supplementary Methods S1-S4**); only their status ceilings and testable content are summarized here.

16.1. Model variants and comparators

The MNDM can be systematically simplified into five hierarchical variants (Minimal-MNDM, fixed-geometry, no-meta, continuous-rhythm, and extended-geometry) to test whether adaptive geometry, the meta-field, or discrete rhythm-locking are actually necessary for the observed signatures. A variant is accepted only when it improves held-out prediction beyond simpler rhythm/precision models. Full variant definitions and comparison logic are given in **Supplementary Methods S1**.

16.2. Noetic Atlas: geometric and topological readouts

The Noetic Atlas applies four complementary readout streams to MNPS-embedded trajectories: manifold embedding/geodesic flow, topological data analysis (Mapper/persistent homology), recurrence quantification analysis, and metastability/criticality indices. Together they form a geometric portrait of denoising dynamics; full stream definitions are given in **Supplementary Methods S1**.

16.3. Stage-1 composite summary (NDHI)

The Noetic Diffusion Health Index (NDHI) is an **exploratory** Stage-1 composite of Atlas readouts, always reported with its component scores and weight-perturbation sensitivity; it is not a clinical biomarker, diagnostic label, or consciousness score. The full construct definition and reporting requirements are given in **Supplementary Methods S1**.

16.4. Truth-known simulator audits: recoverability and reachability

A truth-known synthetic simulator programme, forked from a Slow Branch candidate mechanism for memory linking [34], audits measurement recoverability rather than biology, asking which latent distinctions survive a specified observation map and estimator. The Slow Branch source model shows that replay-dependent slow branch-level accessibility reproduces a joint profile of overlap strengthening, linking gain, perturbation loss, and targeted rescue not matched by nine simpler comparators—a model-**discrimination** result, not evidence that dendrites store memory content. The corrected simulator ladder recovered aggregate reachability calibration under its frozen contract, but its directional Gate R test failed, and a lag-embedded local-operator test was structurally null; neither result validates empirical MNJ estimates or a micro-to-macro neural mechanism. Full recoverability-level definitions and the epoch-operator/ reachability recursion are given in **Supplementary Methods S1** [35].

16.5. Subsystem extensions: CCTG and HELS

Context-Conditional Transition Geometry (CCTG) is a speculative, context- and circuit-dependent replacement for universal TRN-variance-gate language. The Hippocampal-Entorhinal Landscape System (HELS) is orthogonal: it proposes that replay/history may update a context-conditioned landscape F_c. Neither is a Stage-1 result; HELS currently has validated infrastructure but no validated mechanism result, and CCTG must earn support through dedicated reversal/causal tests. Full equations and status are given in **Supplementary Methods S2**.

16.6. NDT-Q: operational content sections

MNPS characterizes a global dynamical regime, not the identity of an experienced or represented content. NDT-Q separates content as an operational readout problem via a regime base π_B, content readout

ρ_b, and content sections, supporting a measurable episode profile $P_t = (c_t, \chi_t, S_t, R_t, V_t)$. This is an explanatory mathematical architecture, not a claim that neural tissue instantiates a literal fibre bundle or that current decoding measures qualia. A first, genuinely blind precursor test now exists in mouse visual cortex (public electrophysiology data): decoders trained only on real, orientation-matched edges generalize to illusory-contour trials beyond a temporal circular-shift null, a non-visual control-region check, and label-shuffled and behavioral-state controls. This targets content-generalization decoding only—no trajectory geometry, content section, or closure signature is computed, no phenomenal report exists in mice, and the result does not itself test or validate NDT-Q, CSQ, or MNPS. The full formalism and evidence ledger are given in **Supplementary Methods S2**.

16.7. Embodied Anchoring Principle (EAP)

EAP states that denoising is conditioned on interoceptive/body state via a self-prior term F_{self} and rhythmic entrainment of $r(t)$; it is a Tier-2 extension intended for a later wave of interoception-focused studies. Current companion evidence supports bounded, condition-level autonomic/MNPS co-modulation in wake and carrier-specific (SO-trough-dominant) autonomic phase-locking in sleep, but not causal body-to-manifold gating, phase-resolved entrainment, or a metabolic consciousness gate (a related hypoglycemia clamp contrast shows a stratified MNPS shift without a confirmed reachability collapse). The full formal integration, neural-substrate hypotheses, phase-landscape model, and Self-Prior Index measurement plan are given in **Supplementary Methods S3**.

16.8. Candidate TRN substrates and CCTG

This records a candidate TRN contribution and a phenomenological simulation mapping without identifying TRN as the biological implementation of NDT's variance schedule: the auditory PV/SOM TRN reanalysis (Zenodo 21774408) supports a dataset-bounded PV-linked contraction of MGB population noise dimensionality, not a full single-unit variance-gating mechanism or identification of $r(t)/\sigma(t)$ with TRN firing. The candidate circuit sketch, simulation ansatz, and CCTG successor hypothesis are given in **Supplementary Methods S2**.

16.9. Related work

NDT's relationship to dynamical-systems/landscape accounts of brain activity, predictive-processing/active-inference frameworks, and integrated- information/network-control approaches is discussed in **Supplementary Methods S4**.

Bibliography

[1] K. Friston, "The free-energy principle: a unified brain theory?," *Nature Reviews Neuroscience*, vol. 11, no. 2, pp. 127–138, 2010, doi: 10.1038/nrn2787.

[2] A. Clark, "Whatever next? Predictive brains, situated agents, and the future of cognitive science," *Behavioral and Brain Sciences*, vol. 36, no. 3, pp. 181–204, 2013, doi: 10.1017/S0140525X12000477.

[3] P. S. Churchland, *Neurophilosophy: Toward a Unified Science of the Mind-Brain*. Cambridge, MA: MIT Press, 1986.

[4] P. S. Churchland, "Neural Representation and Neural Computation," *Philosophical Perspectives*, vol. 4, pp. 341–396, 1990, [Online]. Available: https://patriciachurchland.com/wp-content/uploads/2020/05/1990-Neural-Representation-and-Neural-Computation.pdf

[5] P. S. Churchland, "A Neurophilosophical Slant on Consciousness," in *Progress in Brain Research*, vol. 149, Elsevier, 2005, pp. 285–293. doi: 10.1016/S0079-6123(05)49023-8.

[6] D. C. Dennett, "Real Patterns," *The Journal of Philosophy*, vol. 88, no. 1, pp. 27–51, 1991, doi: 10.2307/2027085.

[7] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-Based Generative Modeling through Stochastic Differential Equations," in *International Conference on Learning Representations*, 2021. doi: 10.48550/arXiv.2011.13456.

[8] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," in *Advances in Neural Information Processing Systems*, 2020, pp. 6840–6851. doi: 10.48550/arXiv.2006.11239.

[9] Y. Song and S. Ermon, "Generative Modeling by Estimating Gradients of the Data Distribution," in *NeurIPS*, 2019. doi: 10.48550/arXiv.1907.05600.

[10] R. Langell, "Reduced Dynamical Capacity Does Not Imply Unconsciousness: Sleep, Propofol, and Epilepsy Dissociate in Transport Capture and Pathological Deformation." [Online]. Available: https://doi.org/10.5281/zenodo.19384063

[11] E. Tognoli and J. A. S. Kelso, "The metastable brain," *Neuron*, vol. 81, no. 1, pp. 35–48, 2014, doi: 10.1016/j.neuron.2013.12.022.

[12] G. Deco, M. L. Kringelbach, V. K. Jirsa, and P. Ritter, "The dynamics of resting fluctuations in the brain: metastability and its dynamical cortical core," *Scientific Reports*, vol. 7, p. 3095, 2017, doi: 10.1038/s41598-017-03073-5.

[13] W. L. Shew and D. Plenz, "The functional benefits of criticality in the cortex," *The Neuroscientist*, vol. 19, no. 1, pp. 88–100, 2013, doi: 10.1177/1073858412445487.

[14] J. Brehmer and K. Cranmer, "Flows for simultaneous manifold learning and density estimation," in *NeurIPS*, 2020. doi: 10.48550/arXiv.2003.13913.

[15] R. T. Canolty and R. T. Knight, "The functional role of cross-frequency coupling," *Trends in Cognitive Sciences*, vol. 14, no. 11, pp. 506–515, 2010, doi: 10.1016/j.tics.2010.09.001.

[16] M. M. Halassa and L. Acsády, "Thalamic inhibition: diverse sources, diverse scales," *Trends in Neurosciences*, vol. 39, no. 10, pp. 680–693, 2016, doi: 10.1016/j.tins.2016.01.002.

[17] M. M. Halassa and S. Kastner, "Thalamic functions in distributed cognitive control," *Nature Neuroscience*, vol. 20, no. 12, pp. 1669–1679, 2017, doi: 10.1038/s41593-017-0020-1.

[18] R. Langell, "Thalamic Reticular Inputs Regulate Population-Level Response Geometry, Not Single-Unit Variability, in the Auditory Thalamus." [Online]. Available: https://doi.org/10.5281/zenodo.21774408

[19] J. E. Lisman and O. Jensen, "The \theta- \gamma neural code," *Neuron*, vol. 77, no. 6, pp. 1002–1016, 2013, doi: 10.1016/j.neuron.2013.03.007.

[20] A. B. L. Tort, R. W. Komorowski, J. R. Manns, N. Kopell, and H. Eichenbaum, "Theta–gamma coupling increases during the learning of item–context associations," *Proceedings of the National Academy of Sciences*, vol. 107, no. 5, pp. 2093–2098, 2009, doi: 10.1073/pnas.0911331106.

[21] B. P. Staresina, J. Niediek, R. Surges, F. Mormann, and V. Borger, “How coupled slow oscillations, spindles and ripples coordinate neuronal processing and communication during human sleep,” *Nature Neuroscience*, vol. 26, pp. 1843–1854, 2023, doi: 10.1038/s41593-023-01381-w.

[22] C.-F. V. Latchoumane, H.-V. V. Ngo, J. Born, and H.-S. Shin, “Thalamic spindles promote memory formation during sleep through triple phase-locking of cortical, thalamic, and hippocampal rhythms,” *Neuron*, vol. 95, pp. 424–435, 2017, doi: 10.1016/j.neuron.2017.06.025.

[23] H.-V. Ngo, J. Fell, and B. P. Staresina, “Sleep spindles mediate hippocampal–neocortical coupling during long-duration ripples,” *eLife*, vol. 9, p. e56655, 2020, doi: 10.7554/eLife.56655.

[24] T. Ng, E. Noh, and R. M. C. Spencer, “Bayesian meta-analysis reveals the mechanistic role of slow oscillation–spindle coupling in sleep-dependent memory consolidation,” *eLife*, vol. 13, p. e101992, 2024, doi: 10.7554/eLife.101992.

[25] R. Langell, “Stratified Meta-Noetic Phase Space: Unmasking Neural Mechanisms beneath the Manifold,” *Zenodo*, Nov. 2025, doi: 10.5281/zenodo.17643473.

[26] R. Langell, “Extreme Neurodynamic Regimes Reveal Meta-Noetic Dissociations.” [Online]. Available: https://doi.org/10.5281/zenodo.18405962

[27] R. Langell, “Manifold Traversability Dissociates Conscious Level from Neural Entropy: An Empirical Evaluation of Noetic Diffusion Theory.” [Online]. Available: https://doi.org/10.5281/zenodo.18141988

[28] R. Langell, “Dynamical Arrest Under Propofol: Reduced Traversability and Meta-Noetic Jacobian Flattening.” [Online]. Available: https://doi.org/10.5281/zenodo.17866987

[29] R. Langell, “Regional Meta-Noetic Phase Space in fMRI: Child-Adult Manifold Dynamics in a Naturalistic Movie.” [Online]. Available: https://doi.org/10.5281/zenodo.17803258

[30] A. Bahramisharif, O. Jensen, J. Jacobs, and J. Lisman, “Serial representation of items during working memory maintenance at letter-selective cortical sites,” *PLoS Biology*, vol. 16, no. 8, p. e2003805, 2018, doi: 10.1371/journal.pbio.2003805.

[31] L. Palaniyappan, “Towards a geometry of psychopathology: a manifold perspective,” *Schizophrenia Bulletin*, vol. 47, no. 3, pp. 651–653, 2021, doi: 10.1093/schbul/sbab022.

[32] C. Sanz, Y. Perl, and R. Carhart-Harris, “Psychedelics, network geometry, and the relaxation of topological constraints,” *Trends in Neurosciences*, vol. 46, no. 8, pp. 608–622, 2023, doi: 10.1016/j.tins.2023.04.005.

[33] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, “Score-Based Generative Modeling through Stochastic Differential Equations,” *ICLR*, 2021.

[34] R. Langell, “Slow Branch-Level Accessibility as a Structural Constraint on Memory Linking.” [Online]. Available: https://doi.org/10.5281/zenodo.20813821

[35] R. Langell, “Separable Levels of Observability in a Truth-Known Memory Simulator.” [Online]. Available: https://doi.org/10.5281/zenodo.21823910